

Landmark Analysis at the 25-Year Landmark Point

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Abstract—This statistical primer presents the landmark analysis method, exploring its appropriate use and interpretation while recognizing its limitations. This observational method is used for comparing time-to-event outcome between groups determined during study follow-up. The goal of the landmark method is to estimate in an unbiased way the time-to-event probabilities in each group conditional on the group membership of patients at a specific time point, the landmark time. The need that led to its development, the impact of the method, and its pros and cons, along with available alternative approaches, are presented. Simulations explore its performance, using realistic parameters from a recent cardiovascular study. As long as the limitations of the method are recognized and the interpretation of its results clearly reflect their “conditional” nature, landmark analysis, 25 years from its introduction, can still be of value. (*Circ Cardiovasc Qual Outcomes*. 2011;4:363-371.)

Key Words: clinical trials ■ observational studies ■ prognostic factors ■ time-to-event outcome
■ time-varying covariate

The landmark method was introduced in 1983, in the first year of the *Journal of Clinical Oncology (JCO)* by Anderson et al,¹ in an article addressing the issue of bias in the analysis of survival of “responders” and “nonresponders,” a popular analysis in oncology. In the erroneous “naïve” analysis, “responders” and “nonresponders” were compared with regard to survival from the start of the study, although their response status was not determined at baseline but later during follow-up. The landmark method has since been used extensively in medical research to correct for the bias inherent in the analysis of time-to-event outcome between groups determined during study follow-up.

In a recent study, landmark analyses were performed to explore the association of extended clopidogrel use and long-term clinical outcomes of patients receiving drug-eluting stents (DES) and bare-metal stents (BMS) for treatment of coronary artery disease.² The article, considered to have recently popularized the landmark approach in cardiovascular studies, particularly of coronary stents, is used later in this review in simulations as a case example.

The goal of this statistical primer is to present the landmark method and explore its appropriate use and interpretation while recognizing its limitations.

Background: The Problem

The biases involved in comparing time-to-event data for different groups in the case that the group membership for an individual arbitrarily varies during a study have been recognized from the early 1970s.³⁻⁵ Such a situation was identified

in the context of heart transplant data in which the survival of patients receiving heart transplants is compared with that of control subjects. The patients receiving a heart transplant must have at least survived from time of diagnosis to time of treatment, whereas no such requirement is necessary for the control subjects. This “time-to-treatment” bias is identical to the “time-to-response” bias in the comparison of overall survival between “responders” and “nonresponders.”

By ignoring in the “naïve” analysis the fact that the group membership variable is changing with time, bias is introduced by several factors. Using the paradigm of classification by response, which is the index case for which the landmark method was developed, these are the timing of response, the impossibility of identifying a response in patients with shorter follow-up for any reason, and the increasing probability of observing a response with longer follow-up. In short, response group classification is dependent on length of follow-up. As noted, the guaranteed time of follow-up alive up to the time of response evaluation that exists in patients belonging to the “responders” group is a time period that no event could be observed in this group.^{1,6} Consequently, the “responders” are erroneously included in the risk group up to the time of identification of response, leading to an inappropriately favorable survival curve for “responders” and an inappropriately unfavorable survival curve for “nonresponders.”

Inadequate statistical comparisons of survival distributions of “responders” and “nonresponders” were identified in approximately 20% of phase II and phase III clinical trials in a survey of articles that appeared from 1979 to 1981 in

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Cancer and *Cancer Treatment Reports*.⁶ In the first 9 issues published in *Cancer* in 1982, a “naïve” analysis of survival of “responders” and “nonresponders” was mentioned in 15 of the 31 reports on treatment of patients with advanced cancers.¹ Even in 1986, 18 reports presenting analysis of survival by tumor response were found in 1 year’s issues of *JCO*, with only 3 recognizing any possible problems with this analysis.⁷

Description of the Landmark Approach

In the landmark method, a fixed time after the initiation of therapy is selected as a landmark for conducting the analysis of survival by response. Only patients alive at the landmark time are included in the analysis, separated into 2 response categories according to whether they have responded up to that time, that is, the landmark method ignores all responses after the landmark time and all deaths before that time. Evaluation of whether survival from the landmark depends on the patient’s response by the landmark is now made possible, with survival probability estimates and correct statistical tests conditional on the response status at the landmark time of patients who survived up to that time. Kaplan–Meier curves, a familiar graphical display in the medical literature, are used for illustrating the estimated conditional survival probabilities, contributing to the clarity of the method.

The landmark method is simple in its execution, its description, and its graphical presentation and is clear in its message. It applies in general to the classification to any groups formed during follow-up on the basis of “a risk elevating or diminishing event.”⁸ Such situations arise commonly within the clinical trial setting, for example, the classification to “treatment” and “no treatment” in the case of patients receiving a heart transplant, the classification to groups formed on the basis of “permanent study medication discontinuation,” “noncompliance,” or “crossing over to the active treatment group,” the latter being a case of particular interest recently.

The goal of the landmark method is to estimate in an unbiased way the survival probabilities in each group conditional on the group membership of patients at the landmark time. Of note, the method is not attempting to address the lack of randomization in the forming of the groups to be compared on the basis of outcome. This and other limitations of the method are discussed later in this review.

Impact of the Method

In the issue celebrating the 25 years of *JCO*, an editorial by the original authors recorded the accomplishment of the established methodology and alternative methods to almost completely eliminate the inappropriate survival by tumor response analyses from the medical literature.⁹ The recognition of the bias in the “naïve” analysis of survival by tumor response was incorporated in the publication guidelines of major cancer journals, recommending the rejection of such comparisons,^{10–12} whereas the Food and Drug Administration has not accepted a survival advantage in responders as evidence of drug benefit.¹³ Even so, as recently as in 2005, the erroneous analysis by response category is present in a *JCO* article,¹⁴ prompting an editorial by Anderson and Neuberg.¹⁵

The relevance and use of the landmark approach did not diminish with the years from its first presentation, and, if anything, the interest in the method seems to be growing. Based on the *International Scientific Index (ISI)*, from 1984 to 2009, landmark analysis is mentioned and the original article cited in more than 420 papers (in 2008, 26 times). According to *Scopus* on citations after 1996, it is cited 224 times in medical journals spanning all diseases, mostly in oncology, in hematology, and in cardiology, where it is cited 12 times (articles are listed in the online-only Data Supplement, Table S3).

In a search conducted for the purposes of this review, all articles appearing in the journal *Circulation*, with the word “landmark” in their title, abstract, or text, were examined to extract the ones actually using the method. In 2009, the landmark method is used for data analysis twice, with 4 times in the previous 2 years, whereas in all previous issues it had been used only twice, once in 2006 and once in 2002 (articles are listed in the online-only Data Supplement, Table S3). It has been used in at least 2 large, observational studies on late coronary stent thrombosis reported in 2007¹⁶ and in a prospective cohort study assessing the relationship between implantable cardioverter-defibrillator therapy and mortality.¹⁷ Hence, the method appears to be of particular and recent interest in cardiovascular research.

In addition, in an identical search of recent articles appearing from 2004 to 2009 in 3 major medical journals, a total of 6, 2, and 9 studies reported in *The Journal of the American Medical Association*, *The Lancet*, and *The New England Journal of Medicine*, respectively, used the landmark method for analysis. From the 6 studies appearing in *The Journal of the American Medical Association*, 5 were on cardiology subjects, including the 2007 article by Eisenstein et al.²

A recent report evaluating the effect of coronary angiography as a time-dependent variable on outcome used an interesting approach and graphical illustration with “repeated” landmark analyses in sequential 6-hour time intervals.¹⁸ Time from hospitalization to angiography was broken into 6-hour time periods. Odds ratios and 95% confidence intervals of delayed outcome (30-day death/myocardial infarction and in-hospital major bleeding) for each 6-hour time period were estimated from logistic regression models and presented graphically for up to 48 hours from hospital admission. The subset of patients without an outcome before the end of each 6-hour period was included in the corresponding landmark analysis. For each period, patients who had angiography within this period were compared with those who had angiography any time later during the index hospitalization or never had angiography.

Limitations

Even in the seminal landmark article, the arbitrary selection of the landmark time and the omission of events occurring earlier to the landmark were among the recognized disadvantages of the method.¹ The method characteristics, advantages, and limitations, along with recommendations on how to address them, are summarized in the Table.

Table. Characteristics of the Landmark Method

Aim	Time-to-event probability estimation, HR estimation and statistical tests, conditional on the group classification at a fixed time point (landmark time)		
Method Characteristics	Observational method		
	Risk group: Only patients event-free at the landmark time Consequence: Ignores all events occurring before the landmark time		
	Group membership: Definition at fixed time point Consequence: Ignores change in group membership after the landmark time		
	Pros		
	Clarity of method: Simple execution, description and graphical illustration via Kaplan-Meier curves		
	Unbiased estimation of conditional time-to-event probabilities		
	Correct conditional statistical tests		
	Cons	Consequences	Possible Remedies
	Dependence on choice of landmark	Omission of time-to-event distribution prior to landmark time	
		Possibility of data-driven results	A priori selection based on clinically significant natural time Sensitivity analysis Use of alternative methods (time-varying Cox models)
Conditional HR estimates: A different target at each landmark time Lack of generalization		Sensitivity analysis Recognition that probability estimate is conditional on the risk set and group definition at the specific fixed timepoint	
Early landmark: Misclassification at longer follow-up		Use of alternative methods (time-varying Cox models)	
Late landmark: Omission of high proportion of events with corresponding loss of power		Avoid generalization to other landmark points or to the overall study course	
Lack of the randomization property in the forming of groups		Nonrandomized comparison of outcome All common caveats of observational methods	Only partial remedy exists by adjusting for known confounders and/or using in parallel other methods (eg, IPW) Use of sensitivity analysis in IPW, measuring how much unmeasured confounding would refute the findings Avoid misinterpretation of significant effects as cause-effect relationships

Choice of Landmark

Omission of Time-to-Event Distribution Before Landmark Time

Data-Driven Results

To safeguard against the danger of a data-driven decision, for example, to avoid additional biases caused by the choice of landmark, the landmark should be selected a priori, based on some clinically significant natural time before the start of data analysis.¹ Alternative methods could be used or a sensitivity analysis could be performed to assess the impact of landmark choice on the analysis results.^{19,20}

As an example of a “natural time” landmark defined a priori, a landmark analysis examined the adverse event rates for patients who had not died or had an ST-segment elevation–myocardial infarction during the first 3 months after stent placement.²¹ In this study, the 3-month landmark was chosen a priori because this was the duration of dual antiplatelet therapy that was recommended in 2003 for patients receiving a sirolimus-eluting stent.

Conditional Hazard Ratio Estimates: A Different Target at Each Landmark Time

The importance of the omission of early events, especially in the case of nonconstant hazards, is illustrated in a study exploring the effect of older and fresh thrombus on risk of death.²² In this study, landmark is specifically used as a means to explore in separate analyses the early and late mortality rates. The information for the first 14 days is available and can be analyzed without the usual biases in the presence of a time-varying group membership. The cumulative Kaplan–Meier estimate of all cause mortality at 4 years (118 events) was significantly higher in the patients with older thrombus. At the landmark analysis at 14 days, the 4-year mortality did not differ significantly between the 2 groups, whereas it did for the first 14 days. The conclusion states that the difference in mortality occurs primarily within weeks after primary percutaneous coronary intervention and is sustained over time. In this specific example, performing only the landmark analysis at 14 days, by omitting 34 early

events along with the high and differential early event rate in the 2 groups, would have been misleading. In such cases that the earlier than the landmark point survival history is different, a fact that cannot generally be verified, as in this study, the omission could lead to possible misrepresentation of the overall survival time model. This limitation of the landmark method is later illustrated through simulations using as a case study the 2007 article by Eisenstein et al.²

The conditional property of a landmark analysis makes it difficult if not inappropriate to generalize its results. It is recommended that along with the presentation of landmark analyses, a sensitivity analysis including data before the landmark should also be provided. In the case that earlier data are not available, simulated data under different scenarios could be used instead.

Misclassification and Loss of Power

The landmark method is clearly most powerful when the “risk altering intervening event” occurs comparatively early and the outcomes of interest are not particularly common at this early study period. In this case, when using an early landmark time point, both the misclassification with longer follow-up and the loss of power would be unimportant. Thus, for example, the landmark analysis is a recognized useful tool for removing guarantee-time bias in the cases that response to treatment occurs early after starting treatment (eg, advanced colorectal cancer), whereas in cases in which response to treatment occurs over an extended period of time (eg, chronic leukemia), use of the landmark method is not indicated.¹⁹

Alternative Method

To retain the early outcome event information, when group membership is defined during follow-up, the alternative approach of fitting a time-varying Cox model could be used to compare time-to-event outcome between the groups.^{5,23} In the time-varying Cox model, group membership is changing from “no risk altering event” to “risk altering event” at the time the intervening event occurs, and thus the arbitrary choice of the appropriate landmark time is not of concern anymore. For example, to explore the effect of coronary angiography to adverse outcome,¹⁸ during the period up to coronary angiography, all patients are considered to belong in the “no angiography” group and thus no comparison is taking place at event times before angiography is documented. Time for all patients is counted from “on-study” or “randomization” and thus all study information is used. This is in contrast to the landmark method, in which all outcome events occurring earlier to the landmark time are excluded. A graphical illustration of the association between a time-varying covariate and outcome has been attempted through an extended Kaplan–Meier estimator.^{8,24}

Lack of the Randomization Property

In any situation, the landmark method is not a panacea—far from it. The biggest limitation of the method stems from the fact that it is used to compare time-to-event outcome between groups not formed on the basis of a random procedure and thus it carries all the caveats of reaching conclusions based on an observational study instead of a controlled experiment. The group membership property is confounded with the

patient characteristics. In the case of “responders,” response is possibly just a marker that selects the good prognosis patients.^{1,9} Because of the breakdown of the randomization property, the significantly longer survival in one group could be attributed to the better prognosis characteristics of the members of this group instead of the property of “group membership.”

It should be recognized that in cases that landmark analysis results to a significant differential effect between groups as defined by the time of landmark, it can only claim “association” and not a “cause-effect” relationship between benefit and group membership. This is consistent with inference within the usual “observational study” framework with the group membership effect corresponding to the “treatment effect.” In this context, it is generally close to impossible to disentangle the effect of a “treatment” from the possibly confounding effects of the underlying factors that lead to “treatment allocation.” Nevertheless, this limitation of the landmark analysis is well known and it is shared by the ad hoc method suggested by Mantel and Byar⁵ or the time-varying covariate Cox model.²³

Inclusion of prognostic variables in the analyses (multivariate modeling) can be used to adjust for known prognostic factors, but the confounding effect cannot unequivocally be remedied, and the possibility of the existence of other factors not accounted for cannot be excluded. Other methods are suggested in this setting.^{25,26} Landmark analysis, even after adjustment, should be recognized and presented as observational data analysis, subject to the limitations of residual confounding, even when conducted on randomized trial data.

Alternative Method

When exploring the association of time to coronary angiography to adverse outcome,¹⁸ the inverse probability weighted method²⁵ was used in parallel to the landmark analysis. In the inverse probability weighted method,²⁵ appropriately used in this context, first the probability of receiving treatment at a particular time is modeled on the basis of measured confounders, accounting for treatment-censoring events, and then the outcomes are weighted by the inverse of the estimated probabilities. The conclusions of the landmark analysis were limited both by their conditionality on angiography status in the subset of patients who are alive and free of myocardial infarction at 6 hours after hospital arrival and by the lack of randomization to time to angiography.²⁷

Even in the inverse probability weighted method, the attempt to correctly account for the confounding factors, apart from sensitivity analysis exploration, is valuable only as far as one’s confidence that all confounding factors are accounted for, that is, under the “no unmeasured confounders” assumption.²⁵ Although analytic adjustments can be used to make the treatment groups more comparable, because of the inability to rule out unmeasured confounders, a sensitivity analysis that measures how much unmeasured confounding must exist to refute the findings should be performed.²⁸ It is an accepted fact of life that randomization cannot be substituted by any analytic method, no matter how elegant.

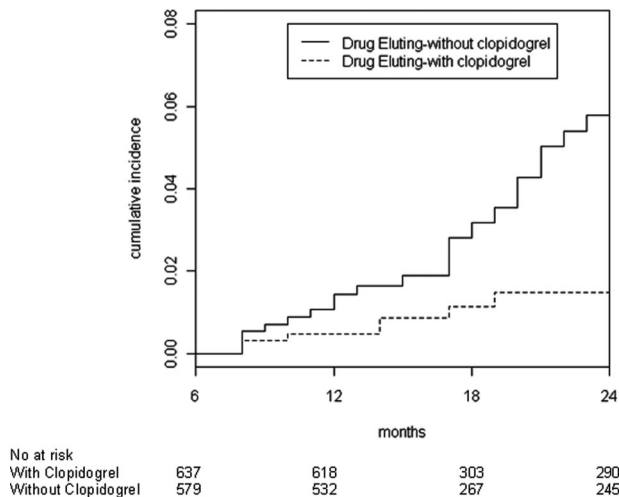


Figure 1. Unadjusted cumulative mortality rates using the 6-month landmark analysis, based on the DES data extracted from the article by Eisenstein et al.²

Exploration of Performance of the Landmark Approach

Simulations, run in the R package (R Development Core Team, 2005),²⁹ explore the limitations of the landmark method and the influence to the results of (1) ignoring the time-to-event distribution before landmark time and (2) defining a time-invariant group membership at the time of the landmark.

More specifically, the simulations illustrate (1) the fact that landmark analysis produces conditional hazard ratio (HR) estimates representing a different target than the unconditional or conditional to another landmark point HR (case 1 and in the online-only Data Supplement, case S1), (2) the effect of misclassification (case 2), and (3) the loss in statistical power (in the online-only Data Supplement, case S2).

Case 1: Conditional HR Estimates: A Different Target at Each Landmark Time

The observational study used here as a case example reported mortality data on patients receiving either a DES or a BMS.² Long-term clinical outcome (up to 24 months) on event-free patients (no death, myocardial infarction, or revascularization) at 6- and 12-month follow-up after initial percutaneous coronary intervention was compared between those receiving clopidogrel and those not receiving clopidogrel. A reduced risk for death was reported for patients with DES with extended use of clopidogrel as defined at the 6-month follow-up (unadjusted $P=0.004$). Inverse probability-weighted-adjusted estimates were also produced ($P=0.03$). In the simplified scenario presented here, no attempt is made to adjust for confounding.

Results for DES are presented here (BMS; in online-only Data Supplement, case S1), using the published data on the 6-month landmark analysis. All mortality and censoring history beyond 6 months is derived directly from the information presented in the report, with HR estimates used for the simulations for DES of 0.29 ($P=0.0049$, Figure 1).

“Extended use” at the time of the study would be defined more appropriately at 3 months instead of at 6 months; the use

beyond the contemporary clinical practice guidelines is of interest, but information on clopidogrel use was not available at 3 months (personal communication, Drs Eisenstein and Anstrom).

In the simulations, “extended use” is defined at 3 months, illustrating the potential effect of the choice of an earlier landmark time (3 months) on the “extended use” published results, based on the 6-month landmark analysis, under different scenarios for the survival distributions and corresponding HRs at this earlier time interval. A Cox model is used for comparing survival. Each simulation experiment is run 10 000 times.

Published Data

At baseline, 1501 patients received DES, and, according to clinical practice, from 3 to 6 months from initial procedure, about 50% of the patients would have discontinued clopidogrel. The 6-month landmark analysis included 1216 event-free patients and a total of 28 deaths (7 with clopidogrel and 21 without), with 48% of the patients not receiving clopidogrel. Sixty-two deaths occurred before the 6-month time point and thus were excluded.²

Simulated Data

Risk of death is assumed constant for the initial 6-month period, and events are taken into account if they occur between 3 and 6 months (on average, half of the 62 deaths), whereas 48% of the patients are assumed to have discontinued clopidogrel (same as reported at 6 months).

Simulation Scenarios for HR From 3 to 6 Months

Four scenarios are explored, with the HR from 3 to 6 months assumed to be (1) $HR=0.29$, the same as observed in the 6-month landmark analysis for DES, (2) $HR=0.86$, the same as observed in the 6-month landmark analysis for BMS, (3) $HR=1$, representing no difference in mortality between patients with clopidogrel and without, and (4) $HR=1.35$, or $1/HR=0.74$, representing a 26% reduction in risk for death of the opposite direction. In each simulation scenario, the history from 6 to 24 months is fixed as derived from the data presented in the manuscript (Figure 1).

The Kaplan–Meier survival curves, under the 4 different scenarios, for the “clopidogrel” and the “without clopidogrel” groups for 1 random realization of the corresponding 10 000 simulated studies, are illustrated in Figure 2, pointing out how identical survival during the 6-month landmark analysis (as reported in the published study) can be produced by totally different survival history up to 6 months.

Results

In the simulations, not surprisingly, as HR from 3 to 6 months progressively takes values further away from $HR=0.29$, a reduced risk for death with extended use of clopidogrel is detected in 79%, 62%, and finally in only 25% of the simulation runs for the landmark analysis at 3 months (Figure 3A and 3B). The online-only Data Supplement material, labeled “Sample Data Set,” contains the sample data set used and the corresponding annotated software code in R.

The fact that only 28 deaths were observed after the 6-month landmark point, whereas 62 that occurred earlier were excluded, explains the impact on the results seen here

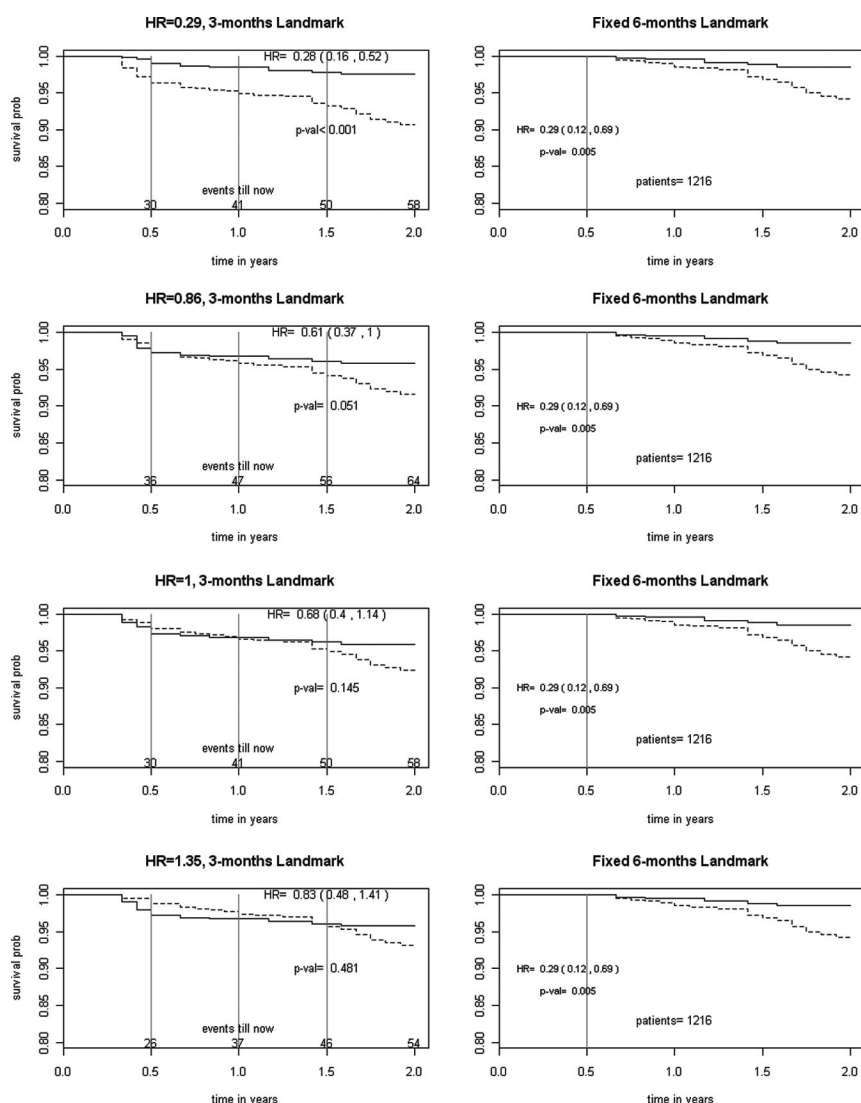


Figure 2. Kaplan-Meier survival plots for 1 random realization from 10 000 simulated studies on DES. Left, Three-month landmark analysis; right, 6-month landmark analysis, fixed on the basis of data presented in Eisenstein et al.² From top to bottom: For the period 3 to 6 months, under exponential distributions for 10 000 simulations, assumed HR=0.29, 0.86, 1.0, and 1.35; 3-month landmark analysis realization, HR=0.28, 0.61, 0.68, and 0.83, respectively. Estimated HRs with 95% confidence intervals and *P*-values are presented inside each panel.

under the different assumed scenarios for approximately half of the omitted deaths. Of note, the impact becomes larger when in the simulation scenarios a higher proportion of patients (than the 52% used), had continued use of clopidogrel during the time period 3 to 6 months (eg, for a 56% proportion and HR=1 from 3 to 6 months, clopidogrel benefit is detected only half of the time).

Remarks

If clinical interest lies on the “extended use” of clopidogrel at 3 months, plausible scenarios on the omitted survival and clopidogrel use in the time period 3 to 6 months could lead to different HR estimates and corresponding conclusions regarding the benefit of “extended use.” It is critical to recognize that the 6-month landmark analysis is landmark time specific and should not be directly generalized to any other time point of interest.

Case 2: Misclassification at Longer Follow-Up

In the simulated scenarios, limited to survival data from 6 to 24 months as in the publication, clopidogrel use is treated as a time-varying covariate, changing over the course of the

follow-up. In this application of interest, the 6-month landmark method is compared with the time-varying Cox model approach. Each simulation experiment is run 10 000 times. Results are presented for DES.

Published Data

According to the published results, clopidogrel use had diminished among patients who reported using it at 6 months (from 100% to 55.2%), whereas in the same period, it increased among patients who reported no use of clopidogrel at 6 months (from 0% to 14.5%). This information could not be taken into account at a landmark analysis because “extended use” of clopidogrel is defined as reported at the specific time point of 6 months.²

Simulated Data

Information on clopidogrel use, available from the publication at 6, 12, and 24 months, is incorporated in the analysis (random switching of groups within each 6-month period, 6 to 12, 12 to 18, 18 to 24; fixed total number of switches). A time-varying covariate Cox model using all study information after the 6-month time point was fitted to the data.

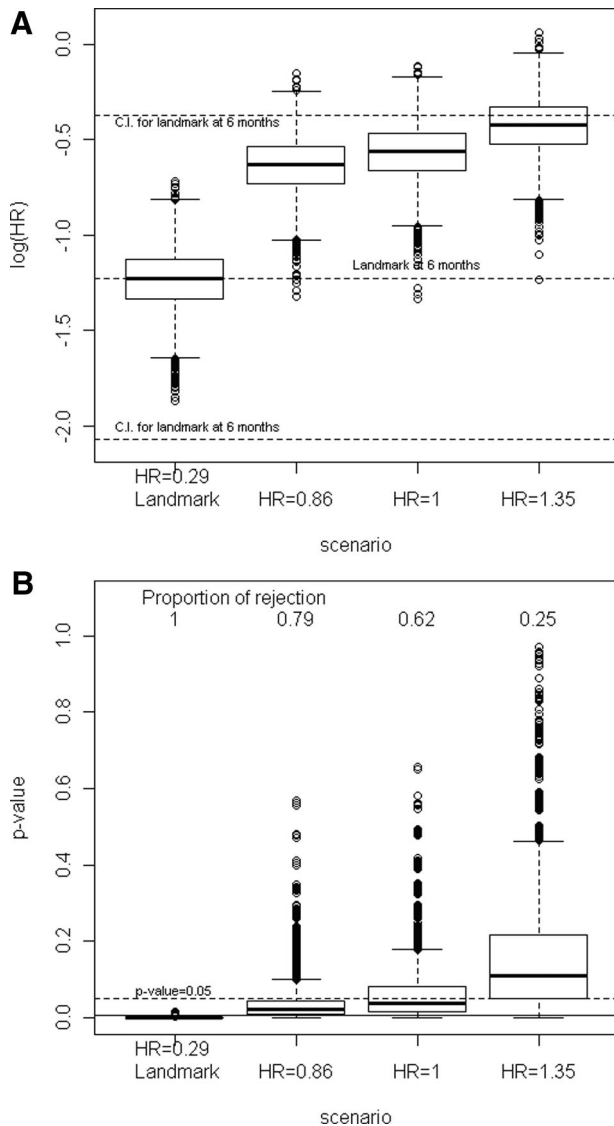


Figure 3. **A**, Box plots of estimated log(HRs). **B**, Box plots of P-values. Three-month landmark analysis for DES; 10 000 simulated studies, with 48% of patients “without clopidogrel.” For the period of 3 to 6 months, assuming under exponential distributions (left to right) HR=0.29, HR=0.86, HR=1, and HR=1.35. The estimated log(HR) with corresponding 95% confidence interval according to the 6-month landmark based on the data in Eisenstein et al.² are presented in **A**. Proportion of simulated studies rejecting the null are presented in **B**.

Simulation Scenarios for Switching From 6 to 24 Months

Two scenarios are explored, each with a different proportion of patients switching during the periods 12 to 18 months and 18 to 24 months (scenario A: 50% in each period; scenario B: 70% from 12 to 18 months). In each simulation scenario, all other aspects of the follow-up history from 6 to 24 months are fixed, based on the data as presented in the manuscript.

Results

Under both scenarios, a reduced risk for death with “extended use” of clopidogrel is detected in only 46%, and 47% of the simulation runs (Figure 4A and 4B). In other words, taking into account a random switching between groups on the basis of the actual reported numbers in the report leads to the

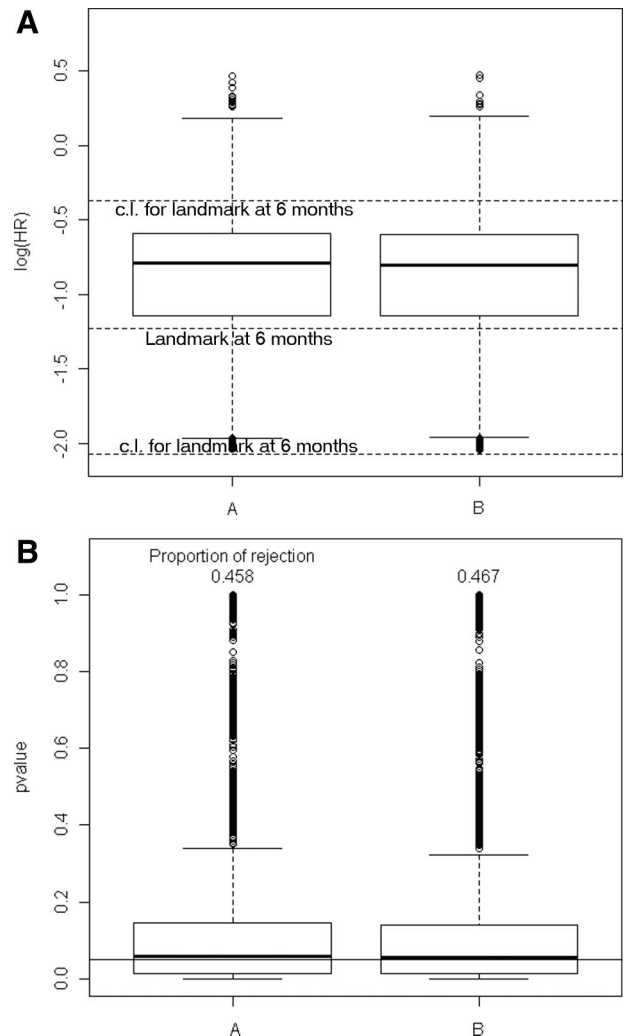


Figure 4. **A**, Box plots of estimated log(HRs). **B**, P-values. Time-varying covariate Cox model is for the period 6 to 24 months. DES, 10 000 simulated studies. Scenarios **A** and **B**: The estimated log(HR) with corresponding 95% confidence interval according to the 6-month landmark, based on the data in Eisenstein et al.² is presented in **A**. Proportion of simulated studies rejecting the null are presented in **B**.

conclusion that a significant benefit of “extended use” of clopidogrel exists in less than half of the simulated cases. Of note, switching to clopidogrel is expected to be more common after a revascularization, a confounding factor that is not taken into account here.

The DES landmark analysis mortality results presented in the report hold only if at least 20 of the 28 observed deaths occur in the time-varying “without clopidogrel” group (Figure 5).

Remarks

If 2 of the 21 deaths reported in the report as observed in the time-invariant “without clopidogrel” group were to occur in patients who switched to clopidogrel at any time beyond 6 months, the benefit of “extended use” of clopidogrel is no longer significant. The effect on the conclusions from possible misclassification is evident.

Concluding Remarks

Landmark analysis is still better than the “naïve” approach for comparing time-to-event outcome in groups, defined on the

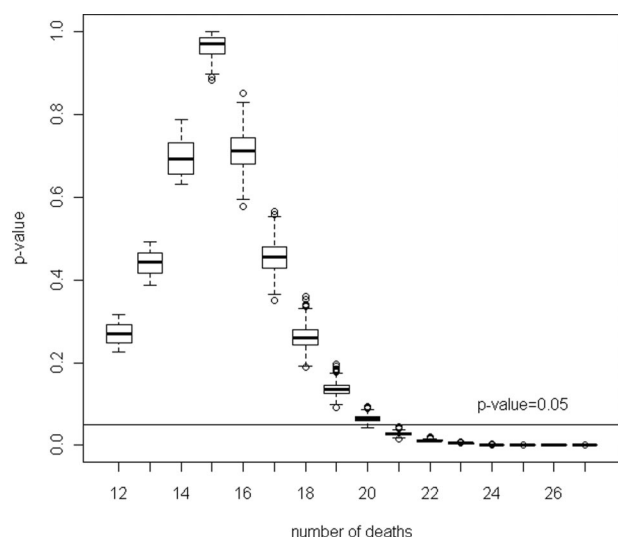


Figure 5. Box plots of P -values from the time-varying covariate Cox model by number of deaths from 6 to 24 months in the “without clopidogrel” group. DES, 10 000 simulated studies under scenario A. A total of 4587 simulated studies with 20 or more deaths led to significant results ($P < 0.05$).

basis of occurrence of an intervening event during follow-up, but in no way compensates for all inherent biases of a nonrandomized comparison. It serves its purpose in estimating time-to-event distributions conditional to the group membership at the landmark time, but this is no substitute to comparing time-to-event distributions between groups with time-varying membership using the full information in a time-varying Cox model.

As the earlier events are omitted and follow-up decreases, not only the power diminishes but also the HR estimated at different landmark points is targeting a different value, which is also different than the HR describing the full survival (time-to-event) history. It is evident that under survival distributions with nonconstant hazards, the landmark analyses results must be recognized as leading possibly to results discrepant to the ones that would be produced using a longer (starting earlier) period of follow-up. The extent of this problem unfortunately cannot be routinely recognized during an actual landmark analysis because it is not generally possible to reconstruct the overall shape and characteristics of the survival distributions so as to make this information available to the investigators. Thus, extra caution should be exercised when interpreting landmark analyses results.

In the particular case that the full course of information is available and the overall and landmark analyses conflict in their conclusions, interpretation is study-specific and should recognize that each analysis answers a different question, with the latter conditional on the specific landmark time point.

The major limitations of this observational method are (1) not accounting for confounders, (2) omitting events that could be important and meaningful even though they occur early, (3) dependence on the choice of landmark point, and (4) dependence on the lack of proportionality and variability of hazards across time that cannot generally be verified under real life conditions. These limitations can unfortunately lead

to landmark time point-specific results that cannot be generalized, substantial loss of power, misleading graphical presentation, and ultimately to wrong conclusions.

In summary, as long as its limitations are recognized and the interpretation of its results clearly reflect their conditional nature, the landmark method, 25 years from its introduction, is still of value (1) in cases that a natural landmark time point is early enough with respect to time-to-event outcome, such that loss of power and misclassification are minimal, (2) because of its simplicity of use and clarity of presentation, along with the more elaborate methods, and (3) in repeated landmark points, in the cases that such a “sensitivity” analysis provides consistent results.

Disclosures

None.

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